

College of Engineering and Technology

Project Work/Internship – Student Hand Book

Project Batch ID

Degree/ program	B.Tech	Specialisation	Computer Science and Engineering Specialisation with AIML
Academic Year	2025-2026 (Even)	Semester	8

Name of student	Register Number	Department	Mobile Number	Email ID
Surya Sathyanaraya nan	RA2211026010562	CINTEL	8838335431	ss7626@srmist.edu.in
Jeevan Babu Murugan	RA2211026010548	CINTEL	8438844636	jm7531@srmist.edu.in

Working Title of the Project:	EdgeSign: An IoT-Enabled Edge-Cloud Hybrid System for Real-Time Sign Language Translation
Project Site / Location	SRM Institute of science and technology Kattankalathur
Name and address of the company/organisation (Applicable for projects with industry or industry support)	SRM IST, Kattankulathur, Chengalpattu District-603203

Supervision Team			
	Supervisor	Co-Supervisor	External Supervisor (If applicable)
Name	Dr.R.Babu		
Designation	Associate Professor		
Department	Cintel		
Campus	Kattankulathur		
Telephone			
E-mail			

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Course Code	21CSP401L/ 21CSP402L	Course Title	Project
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Mission Statement

Problem (or) Product Description: Communication barriers still exist for people with speech and hearing impairments, particularly when interacting with individuals who do not understand sign language. This often leads to dependence on interpreters or written communication. The proposed system, EdgeSign, is an IoT-enabled edge–cloud hybrid solution designed to translate sign language gestures into text and speech in real time. The system captures hand gestures using a camera connected to a Raspberry Pi, extracts hand landmarks using MediaPipe, and processes them using machine learning models. The recognized gestures are converted into meaningful text and further transformed into speech output. By combining edge processing with cloud-based intelligence, the system aims to provide a portable, efficient, and user-friendly communication aid suitable for everyday interactions.

Assumptions and Constraints

Assumptions

- Users perform gestures clearly in front of the camera.
- Adequate lighting conditions are available for gesture detection.
- Stable internet connectivity is available for cloud communication.
- The trained gesture dataset represents common communication gestures.

Constraints

- Raspberry Pi has limited processing power for complex models.
- Recognition accuracy may decrease under poor lighting or cluttered backgrounds.
- Network latency may affect response time during cloud processing.
- Hardware components such as camera modules must be properly configured

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Stakeholders

- Individuals with speech and hearing impairments (primary users).
- Caregivers and family members.
- Educators and rehabilitation centers.
- Developers and researchers working on assistive technologies.
- Organizations promoting accessibility and inclusive communication.

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Division of work and contributors

Time period		Activities or components of the project	Name/Register Number of the Individual Contributor	Names/Register Number of the Joint Contributors
From Date	To Date			
03/01/2026	06/01/2026	Requirement analysis and project planning	Surya sathyanarayanan (RA2211026010562)	Jeevan Baabu Murugan(RA2211026010548)
07/01/2026	10/01/2026	Hardware setup (Raspberry Pi 4, camera), initial configuration	Jeevan Baabu Murugan(RA2211026010548)	Surya sathyanarayanan (RA2211026010562)
11/01/2026	14/01/2026	Installing dependencies (Python, OpenCV, MediaPipe)	Surya sathyanarayanan (RA2211026010562)	Jeevan Baabu Murugan(RA2211026010548)
15/01/2026	18/01/2026	Initial gesture capture testing	Jeevan Baabu Murugan(RA2211026010548)	Surya sathyanarayanan (RA2211026010562)
19/01/2026	22/01/2026	Dataset collection (gesture samples)	Surya sathyanarayanan (RA2211026010562)	Jeevan Baabu Murugan(RA2211026010548)
23/01/2026	26/01/2026	Data preprocessing and cleaning	Jeevan Baabu Murugan(RA2211026010548)	Surya sathyanarayanan (RA2211026010562)
27/01/2026	30/01/2026	Feature extraction using MediaPipe	Surya sathyanarayanan (RA2211026010562)	Jeevan Baabu Murugan(RA2211026010548)
31/01/2026	03/02/2026	Model training (Random Forest)	Jeevan Baabu Murugan(RA2211026010548)	Surya sathyanarayanan (RA2211026010562)
04/02/2026	07/02/2026	Model testing and accuracy tuning	Surya sathyanarayanan (RA2211026010562)	Jeevan Baabu Murugan(RA2211026010548)

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08/02/2026	11/02/2026	WebSocket communication setup	Jeevan Baabu Murugan(RA2211026010548)	Surya sathyanarayanan (RA2211026010562)
12/02/2026	23/02/2026	Sentence formation module	Surya sathyanarayanan (RA2211026010562)	Jeevan Baabu Murugan(RA2211026010548)
24/02/2026	03/03/2026	OCR feature integration	Jeevan Baabu Murugan(RA2211026010548)	Surya sathyanarayanan (RA2211026010562)
04/03/2026	19/03/2026	Speech-to-Text feature integration	Surya sathyanarayanan (RA2211026010562)	Jeevan Baabu Murugan(RA2211026010548)
20/03/2026	27/03/2026	Developer mode (logs & preview)	Jeevan Baabu Murugan(RA2211026010548)	Surya sathyanarayanan (RA2211026010562)

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Time period		Activities or components of the project	Name/Register Number of the Individual Contributor	Names/Register Number of the Joint Contributors
From Date	To Date			
28/03/2026	31/03/2026	System testing and debugging	Surya sathyanarayanan (RA2211026010562)	Jeevan Baabu Murugan(RA2211026010548)
01/04/2026	04/04/2026	Performance optimization	Jeevan Baabu Murugan(RA2211026010548)	Surya sathyanarayanan (RA2211026010562)
05/04/2026	08/04/2026	Final integration of all modules	Surya sathyanarayanan (RA2211026010562)	Jeevan Baabu Murugan(RA2211026010548)
09/04/2026	25/04/2026	Documentation preparation	Jeevan Baabu Murugan(RA2211026010548)	Surya sathyanarayanan (RA2211026010562)
26/04/2026	04/05/2026	Final testing and validation	Surya sathyanarayanan (RA2211026010562)	Jeevan Baabu Murugan(RA2211026010548)

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Summary record of major progress meetings with supervisors (Sprint 1)

Summary record of major progress meetings with supervisors			Working title of dissertation/research project:	
Meeting date & supervisors present	Progress since last meeting	Agreed programme of work and target dates	Other issues, e.g. facilities, supervision, training needs, etc.	Date of next meeting
05/01/2026	Discussed project idea, finalized with problem statement	Complete requirement analysis by 10/01/2026	Need hardware components	12/01/2026
12/01/2026	Hardware all setup started, Raspberry Pi configured	Complete camera integration by 18/01/2026	Minor setup issues	20/01/2026

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20/01/2026	MediaPipe installed, gesture and capture working	Start dataset collection and preprocessing	Need more gesture samples	28/01/2026
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Summary record of major progress meetings with supervisors (SPRINT 2)

Summary record of major progress meetings with supervisors			Working title of dissertation/research project:	
Meeting date & supervisors present	Progress since last meeting	Agreed programme of work and target dates	Other issues, e.g. facilities, supervision, training needs, etc.	Date of next meeting
10/02/2026	Model all training completed, with basic recognition working	Improve accuracy and start text conversion	Accuracy issues in low light	18/02/2026
18/02/2026	Gesture-to-text conversion implemented	Begin WebSocket and cloud integration	Network latency observed	26/02/2026

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26/02/2026	Sentence formation and cloud processing added	Integrate TTS module by 05/03/2026	TTS delay issues	05/03/2026
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Summary record of major progress meetings with supervisors (SPRINT 3)

Summary record of major progress meetings with supervisors			Working title of dissertation/research project:	
Meeting date & supervisors present	Progress since last meeting	Agreed programme of work and target dates	Other issues, e.g. facilities, supervision, training needs, etc.	Date of next meeting
15/03/2026	TTS integrated, audio output working	Improve mobile UI and BLE communication	Minor UI bugs	22/03/2026
22/03/2026	Mobile all application connected and text display working	Add OCR and Speech-to-Text features	Integration complexity	30/03/2026
30/03/2026	OCR, STT, and developer mode added	Final testing and documentation	Performance optimization needed	05/04/2026

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Worksheet / Data collection / Observation etc

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Journal Publication Details

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